Factors Influencing University and High School Students' Reliance on Generative Artificial Intelligence and its Impact on Academic Performance: A Cross-Sectional Survey Informed by the Theory of Planned Behaviour

by

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SELF-DECLARATION

I declare that the report here submitted is original except for the source materials explicitly acknowledged and that this report, or parts of this report have not been previously submitted for the same course or for a different course.

I also acknowledge that I am aware of the Rules on Handling Student Academic Dishonesty and the Regulations of the Student Discipline of the University of Macau.

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Abstract

Artificial Intelligence (AI) is the imitation of human intelligence, involving the development of systems that can perform tasks requiring human-like perceptive abilities, such as learning, reasoning, explaining, and problem-solving. The technical development in this field has been improving and enhancing from day to day, but the progress in researching its impact on students has been constrained due to its importance and contribution to the development of AI at its early stages. This topic is recently emphasized because of its popularity among students, especially being free and available to the public. In this research, a comprehensive overview of the reliance of students on Generative AI (GAI) will be provided, with the research methodology and data analysis methods to show the interpretation of the responses collected from the public.

This cross-sectional study examines the influence of attitudes, subjective norms, and perceived behavioral control from the view of the Theory of Planned Behavior on students' intentions and actual use of GAI. A total of 142 valid responses were sourced through a structured questionnaire and analyzed using descriptive and inferential statistics, including Kendall's Tau correlation tests. The main findings of the study demonstrate that, among students, frequent usage of GAI is positively correlated with the perception that learning new applications is easy, work efficiency is enhanced, and having fun with the technology. Interestingly, support from society plays a stronger role than parental guidance in nurturing GAI skills and ethical consciousness. However, there was no significant correlation between the frequency of GAI use and academic performance, indicating that GAI is ubiquitous across every academic level.

The study reveals how GAI provides personalized learning experiences and efficiency-enhancing prospects for education and, on the other hand, there are also possibilities of such technology being abused. These lessons have different implications for teachers and policymakers regarding how GAI can be used for the good while possibly being curtailed from any negative implications to ensure responsible integration into academic settings.

Introduction

Background of the Study

Artificial Intelligence (AI) Technology has the capability of computational system to complete tasks with the imitation of human intelligence. AI has various critical technologies that can enhance productivity and decision-making in many fields, including education. Generally, AI tools can be divided into the following types: Generative AI (e.g., ChatGPT), Predictive Analytics, Natural Language Processing (e.g., Siri), etc.

The rapid development of Generative AI technology is attracting users around the globe because of its ease of use and ability to generate logically coherent text that meets users' needs. Generative AI

has significantly influenced the current education model, it has demonstrated the transformation of the modern education system and provided students with a new learning approach. The ability of these new technologies to generate text and answer questions with explanations has raised concerns from stakeholders in the education sector. For example, the accuracy of the answers and explanations provided, students' motivation to focus on lectures. But for students, they generally use Generative AI to access personalized learning materials to clarify the knowledge or key points they don't understand. This kind of after class self-learning could be a positive impact instead of negative.

Despite the growing interest in this area, research on the effects of Generative AI on students, particularly in higher education institutions in Macao, a special administration region (SAR) of China, is limited. For these reasons, this project is designed to fill the knowledge gap by exploring this new phenomenon.

Aim of Study

This research focuses specifically on Generative AI, which is a specialised branch of AI designed to produce new content by analysing and learning from existing data. Unlike conversational AI systems, such as online chatbots, which primarily classify or interpret data, Generative AI generates original outputs such as texts, images, videos, etc. In this project, the importance of the research is underscored by its implication for educators, most importantly. As institutions are starting to adopt and use AI tools to enhance the learning quality, it is essential to elucidate both the benefits and challenges of using this new technology.

Despite the growing attention and use of Generative AI in education, there is still a substantial gap in research examining its impact on students. Most existing studies focus on technical aspects or teacher perspectives, often neglecting the direct experiences and attitudes of students. For example, in "The impact of generative AI on higher education learning and teaching: A study of Educator's perspective"(Lee et al., 2024), it highlights that GAI enhances personalised learning, supports creativity, and improves teaching efficiency by the students' exploration of using GAI. On the contrary, concerns regarding the issue of academic integrity and ethical use of GAI persist.

This project will be focusing on students' perspectives, aiming to fill the research gap and provide insights into how Generative AI influences learning motivation and academic performance among students in and/or from Macao SAR of China.

Framework: Theory of Planned Behaviour

The research is guided by the Theory of Planned Behaviour (TPB). The theory states that one's behaviour is shaped by intentions and influenced by three core components: Attitude, Subjective Norms, and Perceived Behaviour Control. This theory provides a comprehensive framework for

understanding how these components affect decision-making processes in different situations.

The three components interact with each other to shape behavioural intentions, which are strong predictors of actual behaviours. TPB suggests that while intentions are crucial for predicting actions, they do not guarantee the behaviours caused by different factors affecting the sense of control over their actions.

This framework is particularly useful for research that involves assessing individual behaviours. Its comprehensive framework integrates multiple psychological factors affecting judgements and behaviours. TPB explicitly incorporates perceived behaviour control, giving an opportunity to account for one's confidence in their ability to perform an action, which is critical in contexts where technology might be used very widely among students. Moreover, its well-defined constructs enable a nuanced understanding of how the components of this theory shape intentions and actions, making it an ideal framework for exploring the complexities of student engagement with Generative AI.

Research Methodology

Survey Questions Design

The survey questions were mapped to the research objectives since they followed the Theory of Planned Behaviour (TPB). Reference was made to earlier studies that looked into students' attitudes toward technology in education, especially those that considered the adoption of AI tools, while research on Generative AI in education formed concrete guidance for framing the questions. Following a literature search, the questions used in this project were adapted from a previous study titled "Factors Influencing University Students' Behavioral Intention to Use Generative Artificial Intelligence: Integrating the Theory of Planned Behavior and AI Literacy" (Wang et al., 2024) which investigates the factors influencing university students' intention to use generative artificial intelligence (GAI), integrating the Theory of Planned Behavior (TPB) and AI literacy, revealing that attitudes mediate the impact of AI literacy and social norms on adoption.The 5-point Likert scale was used as it is widely used in behavioral research for measuring attitude, subjective norm, and perceived behavioral control. The scale provides a wide range of agreement or disagreement, with which a participant may agree - thereby giving much more refined information.

The clarity and relevance of the questions were established by criticizing them internally. Various potential questions were drafted based on the research objectives. A consensus approach was used, which was evaluated by considering clarity, relevance, and alignment with the TPB framework. This process was conducted through consensus. A pilot test was conducted with 10 participants who offered comments on wording, length, and the survey's comprehensibility. Based on these participants' input, ambiguous or redundant questions were dropped or modified.

All team members proficient in both languages participated in translating the English version of the survey into Chinese. Decisions over discrepancies in translation were made by consensus to keep the meaning intact.

Free-text questions were designed and carefully crafted to encourage the participants to share their experiences with Generative AI. For example, open-ended questions like "What specific features of Generative AI do you find most beneficial for your learning process, and why?" could qualify for obtaining qualitative insights through such means. These questions were translated into Chinese along the same lines as the multiple-choice questions to maintain consistency and clarity.

Answers were rated on a numerical code, the 5-point Likert scale ranging from 1 ("Strongly Disagree") to 5 ("Strongly Agree"). The scale was utilized because of its convenience and the straightforwardness for data analysis with a high response sensitivity so that a complete range of perception and attitude can be measured. It also facilitates statistical analysis since it can be analyzed as their aspiration can be transformed into numerical parameters for statistical data analysis.

The project survey was administered through Google Forms due to its user-friendly interface, multi-device compatibility, and data collection feature. Compared to alternative platforms, Google Forms is free of charge and easily accessible to Macao and other respondents, with ease of delivery and access.

Survey Distribution

Survey distribution began on March 3, 2025, and was open for two and a half weeks until March 20, 2025, to take place at a time when most students were likely to be free of academic stress from the deadlines of projects and midterm examinations. This allowed a better quality from the respondents as they doesn’t have to rush for it to save time for revision.

Using a convenience sampling approach, we shared our questionnaire link online through social media including WeChat and Instagram. Online access of the survey had been given to the participants; therefore, no physical form of the survey was printed during the in-person administration. The same process was implemented by all participants without a printed questionnaire by making things less logistically challenging and allowing for analysis of responses directly on our online survey platform.

A major challenge encountered during data collection was the initial slow response rate due to the limited social network. To address this issue, surveys were spread out more diversely into different groups, which is not constrained under the condition of active groups. This approach contributed approximately 15% of responses. Furthermore, to boost the efficiency of data collected, random classmates at different courses’ lectures were requested to complete the survey. This benefits us a wider,

broader and more general insights on the data after research. Nevertheless, the data confidentiality was assured to encourage participation.

Data Cleaning

Thereafter, the data-cleaning procedure was undertaken using systematized coding and formulas in Microsoft Excel. This included removing duplicates, handling missing values, and standardizing responses for consistency. For instance, text responses of n/a or none were recoded to blanks to avoid polluting the dataset with these responses. Any kind of errors would have been ensured during cleaning by being marked for correction due to transparency and reproducibility.

In order to summarize some demographic information, such as years of education, descriptive statistics have been used. Correlation tests (for example, Kendall's Tau) were used for inferential analysis to determine the relationships between certain variables, such as frequencies of AI use in relation to academic performance. Kendall's Tau was used rather than Pearson's r since the data were not normally distributed, which is a prerequisite for Pearson's r. Stacked bar charts were produced to visualize the findings.

R Studio was heavily used in data analysis due to its abilities for inferential statistics and graphics. On the other hand, with an intuitive interface that assists in accessing large datasets and performing advanced analyses, Excel, for its user-friendliness, was also used for computing descriptive statitics. For qualitative data, responses were cleaned using freewordcloudgenerator.com to create word clouds for visualization of important qualitative data from the free-text responses.

Free-text responses were cleaned by removing extraneous characters and standardizing terms. In the present study, a Python script was utilized for the tokenization of the text and the generation of a word cloud representing the most common themes and keywords.

Approval

Approval for the study was given by the Honours College (HC) of the University of Macau following submission of our project proposal. The proposal was assessed and approved by the HC Programme Coordinator about ethicality and institutional context. Upon acceptance on 14 February 2025, the data collection process was allowed to be initiated.

Data Analysis

Data of Respondents

A total of 149 responses were collected between 3rd March and 20th March. Of these, 7 of the participants disagreed their information to be collected, as a result, their responses were not included in further analysis. The remaining 142 responses were considered valid and used for further analysis.

The majority of the responses were obtained from students in tertiary education, and most of them were students studying at the University of Macao. Furthermore, half of the respondents were majoring in Science or Business-related fields. The distribution of students from different area of studies were shown in Figure 1.

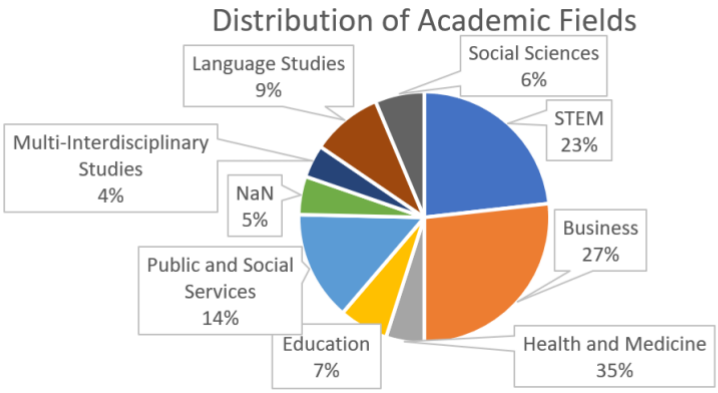


Fig. 1. Distribution of students from different academic fields

With GAI tools for studies, more than 70% of students report that they are being used frequently by the students. There were slight differences in GAI usage frequencies between disciplines. Among the disciplines, probably the highest adoption rates were noted among STEM (23%) and Business (27%) students, who together represent around 50% of the respondents, with almost 80% using GAI tools once a week at least. At 35% advertisement, there were Health and Medicine students next, stating a 75% regular use. On the other hand, Language Studies (9%) and Social Sciences (6%) students were at the low end of usage of GAI, reporting that about 65% of them frequently use GAI. Surprisingly, there is no significant correlation between the frequency of GAI usage and academic performance. This suggests that most of the students rely on GAI in their studies, regardless of all degrees of academic excellence.

Surprisingly, there's no correlation between the frequency of GAI usage and academic performance. A hypothesis could be made that the vast majority of students are using GAI for academics, no matter about their academic excellence.

Factors related to AI literacy that were correlated with frequency of GAI usage

Our correlation analyses showed frequency of GAI usage was positively correlated with some aspects of literacy in AI usage, including finding new GAI applications easy to learn (tau = 0.190, p = 0.002), finding their work efficiency enhanced by using GAI (tau = 0.221, p = 0.0038), finding the support from peers and teachers could positively influence their learning in GAI manipulation (tau = 0.17 and 0.156, p = 0.023 and 0.035 respectively), finding the usage of GAI enjoyable and interesting (tau = 0.24 and 0.218, p = 0.0013 and 0.0028 respectively). Also, there was a tendency that to keep using GAI to solve daily life problems (tau = 0.267, p = 0.0003).

Table 1. Factors related to AI literacy that were correlated with frequency of GAI usage

	Kendall's tau	p-value
l can proficiently use Al applications (e.g. ChatGPT or Sora) or products to assist me in my daily tasks.	0.3726746	6.034e-07
I usually find it easy to learn how to use new GAl applications or products.	0.2990991	5.341e-05
I can use GAl applications or products to enhance my work efficiency.	0.2210872	0.003816
I can evaluate the functionality and limitations of GAl applications (e.g. ChatGPT) or products after using them for a period.	0.2775051	0.00001655
I can choose the appropriate solution from the various solutions provided by GAl-related applications and products.	0.258527	0.0004848
I can select the most suitable GAl application or product for various specific tasks.	0.3170656	1.694e-05
My classmates believe it is necessary to learn how to use GAI technology.	0.1702611	0.02289
My teachers believe it is necessary to learn how to use GAl technology.	0.1559252	0.03348
Learning GAl-related technologies is relatively easy for me.	0.1905459	0.009975

Applying GAl technology to assist me in work and study is relatively easy for me	0.2780602	0.0002076
Applying GAl technology to solve problems in daily life is relatively easy for me.	0.2022968	0.006836
I think it is very wise to apply GAl technology to solve problems in daily life.	0.1852035	0.01355
I find using GAI technology enjoyable.	0.2396866	0.001357
I find using GAI technology for calculations very interesting.	0.2185185	0.002788
I greatly enjoy the convenience brought by applying artificial intelligence technology.	0.2772705	0.0002541
I will continue to keep an eye on the progress of GAl-related technologies.	0.2777751	0.000206
I will regularly update the latest GAl-related applications.	0.3395118	4.47e-06
I plan to use GAl to help me learn and work now and in the future	0.350534	2.996e-06
I will continue to apply GAI technology to solve problems I encounter in my life.	0.2666311	0.0003633

There was a stronger positive correlation found between the GAI usage frequency and AI literacy, including a proficient manipulation of AI application (tau = 0.270, $p \approx 0$), regularly update GAI-related applications (tau = 0.340, $p \approx 0$), and the increasing willingness to use GAI for both learning and working (tau = 0.280, $p \approx 0$).

From the above statistics, it could be precisely spoken that the frequency of AI usage would directly shape their reliance on GAI. The more frequent they use AI tools, the more experience of efficient use of AI will be gained, resulting in an enhancement in efficiency when integrating with work. From their successful experiences that AI could complete tons of workload within a short period of time, and most importantly, without their own efforts. They enjoyed the final result and would keep using and investigating how they can better use AI. This will be a cyclical process without an end, eventually might lead to over-reliance on GAI technology.

The role of support in GAI usage characteristics

Support from parents

From the correlation analyses, as shown in Table 2 parental support for learning to use GAI only exhibited a weak positive correlation. One of the results from the correlation analysis was that higher support from parents was more likely to keep using AI to solve problems (tau = 0.291, $p \approx 0$).

Table 2. Correlation between support from parents and GAI usage characteristics

	Kendall's tau	p-value
I can use GAl applications or products to enhance my work efficiency.	0.1943867	0.008793
I can evaluate the functionality and limitations of GAl applications (e.g. ChatGPT) or products after using them for a period.	0.1429378	0.04568
I can choose the appropriate solution from the various solutions provided by GAl-related applications and products.	0.1798701	0.01241
When using GAl applications or products, I am vigilant about privacy and information security issues	0.2005835	0.004765

I am always vigilant about the misuse of GAI technology.	0.215151	0.0004105
When using GAl applications or products, I always adhere to ethical principles	0.228947	0.001633
Applying GAl technology to assist me in work and study is relatively easy for me	0.1945815	0.007499
Applying GAl technology to solve problems in daily life is relatively easy for me.	0.2263856	0.001823
I find using GAI technology enjoyable.	0.1831407	0.01169
I greatly enjoy the convenience brought by applying artificial intelligence technology.	0.2307471	0.001715
I will continue to keep an eye on the progress of GAl-related technologies.	0.2216762	0.002282
I will regularly update the latest GAl-related applications.	0.2225394	0.00195
I plan to use GAl to help me learn and work now and in the future	0.258519	0.0003877
I will continue to apply GAI technology to solve problems I encounter in my life.	0.2911709	6.061e-05

Support from the society

As shown in Table 3, more and stronger correlations could be found in comparison with the parents' support. The support from the society had a minor but steady influence on their proficiency (tau = 0.284, p ≈ 0). A slight tendency was also observed that societal support would encourage vigilance against GAI's misinformation (tau = 0.21, p = 0.003) and misuse (tau = 0.186, p = 0.009). It could also be found that they would stay informed (tau = 0.28, p ≈ 0) and consider having continued engagement in the future (tau = 0.271, p = 0.0002).

Table 3. Correlation between support from society and GAI usage characteristics

	Kendall's tau	p-value
I can proficiently use Al applications (e.g. ChatGPT or Sora) or products to assist me in my daily tasks.	0.2849756	9.209e-05
I usually find it easy to learn how to use new GAl applications or products.	0.2685262	0.0002014
I can use GAl applications or products to enhance my work efficiency.	0.3809653	3.242e-07
I can evaluate the functionality and limitations of GAl applications (e.g. ChatGPT) or products after using them for a period.	0.3108954	1.529e-05
I can choose the appropriate solution from the various solutions provided by GAl-related applications and products.	0.3452239	1.8e-06
I can select the most suitable GAl application or product for various specific tasks.	0.3183969	9.552e-06

When using GAl applications or products, I am vigilant about privacy and information security issues	0.2096325	0.003336
l am always vigilant about the misuse of GAI technology.	0.1863638	0.009193
When using GAl applications or products, I always adhere to ethical principles	0.1599961	0.02851
Learning GAl-related technologies is relatively easy for me.	0.3379925	2.814e-06
Applying GAl technology to assist me in work and study is relatively easy for me	0.397345	5.557e-08
Applying GAl technology to solve problems in daily life is relatively easy for me.	0.3971606	5.274e-08
I think it is very wise to apply GAl technology to solve problems in daily life.	0.354178	1.307e-06
I find using GAI technology enjoyable.	0.3517709	1.446e-06
I find using GAI technology for calculations very interesting.	0.2838489	6.88e-05
I greatly enjoy the convenience brought by applying artificial intelligence technology.	0.3874736	1.615e-07

I will continue to keep an eye on the progress of GAI-related technologies.	0.2908412	6.817e-05
I will regularly update the latest GAI-related applications.	0.2846833	8.056e-05
I plan to use GAI to help me learn and work now and in the future	0.2710698	0.0002141
I will continue to apply GAI technology to solve problems I encounter in my life.	0.3773813	2.32e-07

A more robust relationship could be found during the analysis. For example, a practical improvement of efficiency (tau = 0.38, p ≈ 0) and solution selections (tau = 0.345, p ≈ 0) could be accomplished. Also, it facilitated learning and application (tau = 0.400, p ≈ 0) and also emotional satisfaction (tau = 0.351, p ≈ 0).

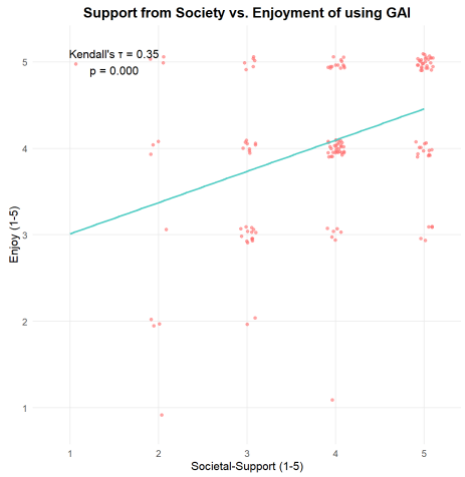


Fig. 2. Correlation of "Societal-Support" with "Enjoy".

From the above data, it could be concluded that societal support could make them feel it is wise to adapt and get used to new modern technologies to assist their lives. They didn't have to feel shy or shame for using it, instead, the social connections between them are also using and learning AI with them, causing them to have a better general overall insight on both the development and updates GAI. As a result, their knowledges and proficiencies in using GAI will gradually increase.

Support from teachers

From the correlation analyses, there were some moderate correlations with the support from teachers towards GAI technology. All these relationships illustrated that teachers' support concentrates on critical application and ethical foresight; It had a stronger relationship with both students' vigilance on the misuse of GAI (tau = 0.333, $p \approx 0$), the formation of the notions about utilizing GAI is a clever approach (tau = 0.406, $p \approx 0$) and utilizing GAI is easy for problem-solving (tau = 0.322, $p \approx 0$), where teachers' support played a role of emphasizing judicious and context-aware tool use, imposing a strategic and ethical impact on students' idea.

Table 4. Correlation between support from teachers and GAI usage characteristics

	Kendall's tau	p-value
I can proficiently use Al applications (e.g. ChatGPT or Sora) or products to assist me in my daily tasks.	0.1442393	0.0457
I usually find it easy to learn how to use new GAl applications or products.	0.2984549	3.035e-05
I can use GAl applications or products to enhance my work efficiency.	0.3322264	6.871e-06
I can evaluate the functionality and limitations of GAl applications (e.g. ChatGPT) or products after using them for a period.	0.2697567	0.0001519
I can choose the appropriate solution from the various solutions provided by GAl-related applications and products.	0.231658	0.001219

I can select the most suitable GAl application or product for various specific tasks.	0.2953545	3.387e-05
When using GAl applications or products, I am vigilant about privacy and information security issues.	0.1800166	0.01095
I am always vigilant about the misuse of GAl technology.	0.3334243	2.544e-06
When using GAl applications or products, I always adhere to ethical principles.	0.1809837	0.01238
Learning GAl-related technologies is relatively easy for me.	0.294489	3.791e-05
Applying GAl technology to assist me in work and study is relatively easy for me	0.293671	5.05e-05
Applying GAl technology to solve problems in daily life is relatively easy for me.	0.3217543	8.564e-06
I think it is very wise to apply GAl technology to solve problems in daily life.	0.4063355	2.093e-08
I find using GAI technology enjoyable.	0.4369473	1.52e-09
I find using GAI technology for calculations very interesting.	0.3613761	3.125e-07

I greatly enjoy the convenience brought by applying artificial intelligence technology.	0.304854	3.168e-05
I will continue to keep an eye on the progress of GAl-related technologies.	0.3352906	3.571e-06
I will regularly update the latest GAl-related applications.	0.3083516	1.624e-05
I plan to use GAl to help me learn and work now and in the future	0.336075	3.6e-06
I will continue to apply GAI technology to solve problems I encounter in my life.	0.4065011	1.869e-08

Meanwhile, their support also contributed to positive correlations with students' agreement to use GAI for learning and working in the future (tau = 0.336, $p \approx 0$), successive application of GAI for solving problems in daily life (tau = 0.407, $p \approx 0$), as well as the willingness of regular updating of GAI technologies (tau = 0.308, $p \approx 0$), these relationships implied that advice from teachers reinforces students' habits for long-term, responsible reliance on GAI tools.

At the same time, teachers linked the use of GAI tools to meaningful outcomes for students while making engagement intrinsically rewarding. This concept could be concluded from the correlation analysis of teachers' support with the extent to which students enjoy utilizing GAI (tau = 0.437, $p \approx 0$), find it interesting and convenient (tau = 0.361 and tau = 0.304, $p \approx 0$ and $p \approx 0$, respectively).

Support from friends or classmates

To be more specific, from peers, there were positive correlations between their support for some aspects of AI literacy and subjective norms. To begin with, it could be observed that peers' support fostered students to master GAI tools more quickly and integrate them into daily work, as it contributed to typical connections with their proficiency in using GAI (tau = 0.334, $p \approx 0$), enhancement of work efficiency (tau = 0.410, $p \approx 0$), and their feeling of ease in using (tau = 0.328, $p \approx 0$), learning (tau = 0.410, $p \approx 0$), applying to work or studies and to resolve problems (tau = 0.414 and tau = 0.407, $p \approx 0$ and $p \approx 0$, respectively) about GAI-related technologies.

Table 5. Correlation between support from friends or classmates and GAI usage characteristics

	Kendall's tau	p-value
I can proficiently use Al applications (e.g. ChatGPT or Sora) or products to assist me in my daily tasks.	0.3336067	5.293e-06
I usually find it easy to learn how to use new GAl applications or products.	0.3278778	7.106e-06
I can use GAl applications or products to enhance my work efficiency.	0.4102844	5.229e-08
I can evaluate the functionality and limitations of GAl applications (e.g. ChatGPT) or products after using them for a period.	0.3604259	7.045e-07
I can choose the appropriate solution from the various solutions provided by GAl-related applications and products.	0.3664972	5.306e-07
I can select the most suitable GAl application or product for various specific tasks.	0.2360183	0.001167
I am always vigilant about the misuse of GAl technology.	0.1534777	0.03381
Learning GAl-related technologies is relatively easy for me.	0.4095698	1.961e-08

Applying GAl technology to assist me in work and study is relatively easy for me	0.414513	2.06e-08
Applying GAl technology to solve problems in daily life is relatively easy for me.	0.4072984	3.365e-08
I think it is very wise to apply GAl technology to solve problems in daily life.	0.3416521	3.877e-06
I find using GAI technology enjoyable.	0.30295	4.032e-05
I find using GAI technology for calculations very interesting.	0.4033665	2.191e-08
I greatly enjoy the convenience brought by applying artificial intelligence technology .	0.4197007	1.975e-08
I will continue to keep an eye on the progress of GAl-related technologies.	0.3465256	2.672e-06
I will regularly update the latest GAl-related applications.	0.3384878	3.518e-06
I plan to use GAl to help me learn and work now and in the future	0.3593499	1.203e-06
I will continue to apply GAI technology to solve problems I encounter in my life.	0.3628035	8.702e-07

In addition, peer interactions had made students feel that using GAI is not only fun and accessible but also that GAI tools are worth long-term clinging to. This was reinforced by the positive relations with the extent to which students enjoy utilizing GAI (tau = 0.303, $p \approx 0$), find it interesting (tau = 0.403, $p \approx 0$) and convenient (tau = 0.420, $p \approx 0$), and willingness to apply GAI for continuous utilization of it (tau = 0.363, $p \approx 0$) or even in work (tau = 0.359, $p \approx 0$) in the future.

Findings

The research describes the complete indication of interaction between university and high school students with GAI tools in the academic work process. The study reveals the usage frequency and external influences in performing behavioural outcomes. Guided by TPB as a theoretical framework, this study highlights the complex relationship of students depending on GAI with various social and educational factors influencing their use of the technology.

The most prominent of these findings is that there exists a significant relationship between the frequency of use of GAI and students' increasing dependence on these tools. More frequent legal use was shown to increase the students' self-reported proficiency in using AI applications, the extent to which they believed their work efficiency improved and the average tendency to include AI in their day-to-day problem-solving lists. This pattern suggests being self-reinforced, whereby the students started off using GAI and mostly have positive experiences of using it and would later need the assistance of GAI for far more complex tasks. While such trends promise practical uses for GAI in increasing productivity, they also raise concerns regarding overdependence: that is, students are learning to make AI-generated solutions more important than their independent thinking.

The external support systems turned out to be an equally important external determinant of student's attitudes and behaviours towards GAI. Peer support, in particular, emerged to have had very strong effects such that students who were actively encouraged by their classmates had higher levels of confidence and efficiency in AI tool usage. Teacher support did follow a different route but by no means one less significant; that was essentially in developing ethical awareness and responsible usage. Educators advocating GAI as an aid in learning invariably helped students assimilate a more strategic and discerning approach to the technology while maximizing its advantages against risks such as misinformation and academic misconduct, in contrast with parents who would have rather opposed claims that parental support has less but measurable effects-such as parental support, encouraging continued use but not proficiency or ethical consideration.

Societal attitudes toward GAI also rode along with the adoption into students' behaviour. The more the society learned about and accepted AI tools, the more the students tended to become eager to know the new developments and integrate GAI into their long-term academic strategies. This kind of societal reinforcement, coupled with peer and teacher influences, would set up the kind of environment in which students felt secure enough to explore and, in the end, turn to GAI for tasks ranging from research to creative problem solving.

Beyond external factors, the main attitudinal changes which now emerge from frequent usage of GAI also have been revealed in the study. Students who found AI tools extremely enjoyable and easy at the most likely consider them as tools without which they cannot imagine studying. This kind of perception is reinforced by emotional satisfaction and ease of usage, which form a feedback loop in

which positive experiences lead to even greater engagement. However, this enthusiasm also had the potential to be a double-edged sword, with some students confessing to employing GAI as a cut-through for activities that might otherwise entail more substantial intellectual toil.

Together, these findings paint a picture of a world in which GAI is increasingly being integrated into scholarly workstreams, both due to its functional utility and the social dynamics of using it. While the technology is clearly improved in efficiency and accessibility, its growing visibility necessitates careful consideration of balancing these advantages against upholding critical thinking and academic honesty. The study calls for formal instruction by instructors and institutions to ensure that GAI is an adjunct to and not a replacement for students' own learning processes. Future research could generalize these findings by studying how different disciplines integrate GAI and how cultural or organizational contexts affect its adoption.

Limitations

Even with its contributions, the current study has several limitations that must be taken into account. Firstly, the sample is largely made up of university students, especially those from the University of Macao, and this might restrict generalizability of results to other groups, for example, professionals or students from other cultural and educational contexts. Secondly, the use of self-reported data brings in biases where participants tend to overreport their ability or underreport difficulties in working with GAI. In addition, while correlation analyses point towards significant relationships, they do not suggest causality, i.e., further experimental or longitudinal research would be needed to determine whether increased use of GAI necessarily leads to dependency.

Another limitation is the local focus on Macao, which may not be representative of global trends in the adoption of AI, especially in countries with alternative technological infrastructures or educational policy. Finally, the study does not thoroughly examine potential negative consequences of over-reliance, such as reduced critical thinking skills or ethical concerns, that would be valuable areas of future research. Overcoming these limitations in future research would provide a clearer image of GAI's long-term impact on work and study habits.

Conclusion

The findings of this study describe how often the utilization of Generative AI (GAI) has an effective impact on the reliance of students on the technology. The higher the usage of GAI tools among students, the more skilled they become, the more productive their work is, and the higher their inclination towards continuing to use AI to address issues in educational and day-to-day life settings. Positive exposure to GAI enhances adoption, creating a self-reinforcing cycle that can result in over-reliance if not balanced by critical consciousness. Peer and teacher pressure is the most potent source of influence among them all; it generates both functional and ethical issues with AI adoption. Parental

support is useful somehow, and social support also helps by making GAI adoption more normal and enhancing efficiency. Studies generally point towards the fact that while GAI has immense learning and productivity benefits, higher integration must be balanced to avoid overdependence and being used to maximum advantage.

Reference

Lee, D., Arnold, M., Srivastava, A., Plastow, K., Strelan, P., Ploeckl, F., Lekkas, D., & Palmer, E. (2024). The impact of generative AI on higher education learning and teaching: A study of educators' perspectives. Computers and Education: Artificial Intelligence, 6, 100221. https://doi.org/https://doi.org/10.1016/j.caeai.2024.100221

Wang, C., Wang, H., Li, Y., Dai, J., Gu, X., & Yu, T. (2024). Factors Influencing University Students' Behavioral Intention to Use Generative Artificial Intelligence: Integrating the Theory of Planned Behavior and AI Literacy. International Journal of Human-Computer Interaction. https://doi.org/10.1080/10447318.2024.2383033